

Article

Effect of Three Kinds of Aloe Emodin Microcapsules Prepared by SDBS, OP-10 and TWEEN-80 Emulsifiers on Antibacterial, Optical and Mechanical Properties of Water-Based Coating for MDF

Ye Zhu ^{1,2}, Nan Huang ^{1,2} and Xiaoxing Yan ^{1,2,*}

¹ Co-Innovation Center of Efficient Processing and Utilization of Forest Resources, Nanjing Forestry University, Nanjing 210037, China; zhuye@njfu.edu.cn (Y.Z.); huangnan@njfu.edu.cn (N.H.)

² College of Furnishings and Industrial Design, Nanjing Forestry University, Nanjing 210037, China

* Correspondence: yanxiaoxing@njfu.edu.cn

Abstract: To investigate an impact of microcapsules on water-based coatings for a medium density fiberboard (MDF), three types of aloe emodin microcapsules prepared at a 3.0% concentration of SDBS, OP-10 and TWEEN-80 emulsifiers were added to the water-based coatings and applied on a MDF surface. Then, three sets of coatings were tested. Antibacterial properties of three groups of coatings on MDF surface increased with the addition of aloe emodin microcapsule content, and antibacterial properties of the water-based coating against *Escherichia coli* in the three groups were slightly superior to that of *Staphylococcus aureus*. The water-based coatings on the MDF surface with SDBS as the emulsifier showed the highest antibacterial rates of 74.1% and 66.0% against *Escherichia coli* and *Staphylococcus aureus*, respectively. The antibacterial rates of 70.0% and 62.8% were achieved for the OP-10 emulsifier group, and 67.0% and 61.9% for the TWEEN-80 emulsifier group, respectively. The aloe emodin microcapsules prepared at a 3.0% concentration of SDBS, OP-10 and TWEEN-80 emulsifiers inhibit bacterial growth and improve the mechanical and optical properties of coatings. It supplies technical references for the utilization of aloe emodin microcapsules to the antibacterial coating on the surface of wooden furniture.

Keywords: microcapsule; antibacterial; emulsifier concentration; mechanical property



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1. Introduction

Wooden materials are widely used in production of wood products and furniture due to their unique grain and texture, easy processing, light weight, and high strength [1,2]. However, the wood tends to shrink and swell with water loss and absorption, which leads to problems such as dimensional instability [3]. A medium density fiberboard (MDF) is widely used as a wooden substrate for furniture and wood products. MDF is susceptible to damage such as decay and mold [4,5] and absorbs moisture from the environment, providing favorable conditions for bacterial growth [6,7]. MDF products are often coated to protect the substrate [8]. Compared to traditional coatings, water-based coatings have advantages such as environmental protection, energy conservation, and low formaldehyde content [9]. Therefore, the water-based coatings are being used more and more widely [10,11]. Over the past few years, much research has been conducted on the modification of water-based coatings with functions such as anti-flash rust [12], anti-static electricity [13], and color change [14]. Coatings with antibacterial properties on the surface of wood products curb the growth of bacteria, thus reducing the density of bacteria [15], achieving the purpose of protecting coatings and optimizing the living environment. Antibacterial agents added to coatings are usually classified as natural, inorganic or organic. In the research findings of enhancing the antibacterial properties of water-based coatings, metal antibacterial agents or metal and natural composites of

antibacterial agents are widely used [16], but little is reported about the natural antibacterial agents. Some natural antibacterial agents suffer from poor processing properties [17,18], unstable chemical properties, and other problems [19]. Therefore, the development of natural antibacterial agents in water-based coatings is limited, making it difficult to produce on a large scale.

Microencapsulation technology has the advantages of changing physical properties and protecting physiological activities of a core material [20,21]. Application and research of the microencapsulation technology for coatings have developed rapidly in recent years. Peng et al. [22] added cellulose to urea–formaldehyde resin to produce a wall material coat tung oil. The tung oil microcapsules enhanced the self-restoration of the water-based coatings on the wood surface. Shahabudin et al. [23] used poly-urea–formaldehyde resin as the wall material and alkyd extracted from palm oil as the core material to prepare thermally stable microcapsules that can be used in self-healing coatings. These findings demonstrate that the microcapsules hold the functions of the core material and offer the coatings a diverse range of functions.

Aloe emodin is an anthraquinone compound [24]. It is a natural organic compound extracted from aloe, and has anti-inflammatory, bactericidal, anti-viral, and other impacts [25–28]. Aloe emodin is also an effective constituent of the antibacterial function [29]. Aloe emodin is a natural antibacterial agent that is extremely sensitive to *Staphylococci* and *Streptococci*, which can effectively destroy the bacteria while being safe, harmless, and non-irritating [30]. In the study of the antibacterial mechanism of aloe emodin, Hu et al. [31] studied the antibacterial activity of a variety of anthraquinone derivatives, including aloe emodin by microcalorimetric technology. The results showed that aloe vera rhodopsin has an inhibitory effect on *Staphylococcus aureus*. In the antibacterial application of aloe emodin, the aloe vera or aloe emodin have been used directly for material modification and antimicrobial tests in recent years. However, studies on the use of aloe emodin into microcapsules and applied to water-based coatings are few [32,33].

In previous research, an optimal preparation process for aloe emodin microcapsules has been demonstrated. The process involved a mole ratio of formaldehyde to urea of 1.2:1, a core-to-wall ratio of 1:15, a reaction temperature of 50 °C, and a stirring speed of 600 rpm for microencapsulation [34]. Three emulsifiers (SDBS, OP-10, and TWEEN-80) at a 3.0% concentration showed better results compared to a 1.0% concentration. Additionally, when compounding the microcapsules with the water-based coatings, it was observed that the antibacterial performance of the water-based coatings was enhanced. When the water-based coatings are applied on the surface of wood products, further verification is required to determine whether the coating with microcapsules still has antibacterial properties and whether the antibacterial properties of the coating can be enhanced. Therefore, three types of emulsifiers with a 3.0% concentration of SDBS, OP-10, and TWEEN-80 were selected to prepare microcapsules. Then, they were added to the water-based coating and coated on an MDF. Then, the influence of emulsifier type and addition level on the performance of the water-based coatings was investigated through the characterization of morphology, chemical composition, antibacterial properties, optical properties, and mechanical properties. The aloe emodin microcapsules not only overcame the defect of dark color and could not be directly added to the coating, but also improved the antibacterial function of water-based coatings with a potential application value.

2. Materials and Methods

2.1. Materials

Table 1 shows the materials that were used in this test. Primer and topcoat were Dulux water-based acrylic varnish, and the size of the MDF was 50 mm × 50 mm × 5 mm (Shangpin Bense Smart Home Co., Ltd., Zaozhuang, China). The diameter of the petri dish was 90 mm. The second-generation type strain ATCC25922 *Escherichia coli* and the second-generation standard strain ACTT6538 *Staphylococcus aureus* were used, both obtained

from the Beijing Microbiological Culture Collection Center, Beijing. The emulsifiers were obtained from Tianjin Beichen District Fangzheng Reagent Factory, Tianjin.

Table 1. List of raw materials.

Material	Molecular Formula	M _w (g/mol)	CAS No.	Concentration (%)
urea	CH ₄ N ₂ O	60.06	57-13-6	99.0
formaldehyde solution	-	-	-	37.0
triethanolamine	C ₆ H ₁₅ NO ₃	149.19	102-71-6	99.9
polyvinyl alcohol	[C ₂ H ₄ O] _n	-	9002-89-5	-
aloe emodin	C ₁₅ H ₁₀ O ₅	270.2369	481-72-1	98.0
citric acid monohydrate	C ₆ H ₁₀ O ₈	210.14	5949-29-1	99.9
anhydrous ethanol	C ₂ H ₆ O	46.07	64-17-5	99.9
waterborne acrylic resin	-	-	9003-01-4	-
<i>Escherichia coli</i>	-	-	-	-
<i>Staphylococcus aureus</i>	-	-	-	-
nutrient agar medium	-	-	-	-
nutritional broth	-	-	-	-
sodium chloride	NaCl	58.4428	7647-14-5	99.5
silicon dioxide	SiO ₂	60.084	14808-60-7	99.5
polyethylene film	-	-	-	-
petri dish	-	-	-	-
sodium dodecylbenzene sulfonate (SDBS)	C ₁₈ H ₂₉ NaO ₃ S	348.48	25155-30-0	99.0
Octylphenol polyoxyethylene ether-10 (OP-10)	ether-(C ₂ H ₄ O) _n ·C ₁₄ H ₂₂ O	602.797	9002-93-1	99.0
Polyoxyethylene dehydrated sorbitan monooleate (TWEEN-80)	C ₂₄ H ₄₄ O ₆	428.60	9005-65-6	99.0

2.2. Preparation Method of the Aloe Emodin Microcapsules

Table 2 presents the emulsifier type and the corresponding HLB values. Microcapsules were prepared based on the parameters specified in Table 3, and “#” is a sample number unit. The materials for the aloe emodin microcapsules are listed in Table 4.

Table 2. Different emulsifier types and HLB value.

Name of Emulsifier	HLB Value	Type of Emulsifier
SDBS	10.6	Anionic type
OP-10	13.4	Non-ionic
TWEEN-80	15.0	Non-ionic

Table 3. Preparation parameters of the test.

Sample (#)	Types of Emulsifiers	Emulsifier Concentration (%)	n (Urea): n (Formaldehyde)	m (Core Material): m (Wall Material)	Temperature (°C)	Stirring Speed (rpm)
1	SDBS	3	1:1.2	1:15	50	600
2	OP-10	3	1:1.2	1:15	50	600
3	TWEEN-80	3	1:1.2	1:15	50	600

Table 4. Material list for aloe emodin microcapsules.

Sample	Urea (g)	Formaldehyde Solution (g)	Wall Material (g)	Polyvinyl Alcohol (g)	Aloe Emodin (g)	Deionized Water (g)	Emulsifier (g)	NaCl (g)	SiO ₂ (g)
1	10.00	16.22	16.00	0.10	1.07	227.95	7.05	1.28	1.28
2	10.00	16.22	16.00	0.10	1.07	227.95	7.05	1.28	1.28
3	10.00	16.22	16.00	0.10	1.07	227.95	7.05	1.28	1.28

(1) Preparation of wall materials: A certain amount of urea was used to mix the formaldehyde solution, followed by the addition of triethanolamine to achieve a pH value of 8. Subsequently, 0.1 g of polyvinyl alcohol was added into the mixture. The resulting mixture was then put on a magnetic stirrer apparatus and heated to 80 °C, while it was continuously stirred at 600 rpm for a duration of 1 h, and urea–formaldehyde resin was obtained.

(2) Preparation of core materials: Different emulsifiers were blended with water and stirred evenly. Then, the core materials were added to the mixed solution. The reaction temperature was regulated to 50 °C and the stirring rate to 1000 r/min, making the core material fully emulsify for 45 min and ensuring that the emulsifier was uniformly wrapped on the outside surface of the core material.

(3) Microencapsulation: The urea–formaldehyde resin as wall material was gradually dripped into the core material at a speed of 600 rpm. The citric acid monohydrate was added to regulate the pH value of the solution to 2.5–3.0. Then, an appropriate amount of NaCl and SiO₂ powder was added and the mixture was stirred for 2 h at 50 °C. After a 24 h aging at room temperature, the final product was filtered using ethanol and water. Subsequently, the solid product was dried at 40 °C for 24 h. Then, the obtained powder was aloe emodin microcapsules.

2.3. Preparation of Coatings

The coating process is presented in Figure 1. The process of coated MDF was manual brushing, with two layers of primer and two layers of topcoat. Each layer had a density of 78 g/m², total coating amount was 312 g/m², and coating thickness was about 80 µm. According to the area of the MDF used, a theoretical average total amount of coating per piece of MDF was 0.78 g. However, in actual painting, the actual dosage should take into account losses during brushing, and the actual consumption of coating was 1.5 to 1.8 times of the theoretical coating amount. Therefore, in actual application, the average total amount of coating used for each MDF was 1.4 g. The coating materials for different microcapsule additions are shown in Table 5.

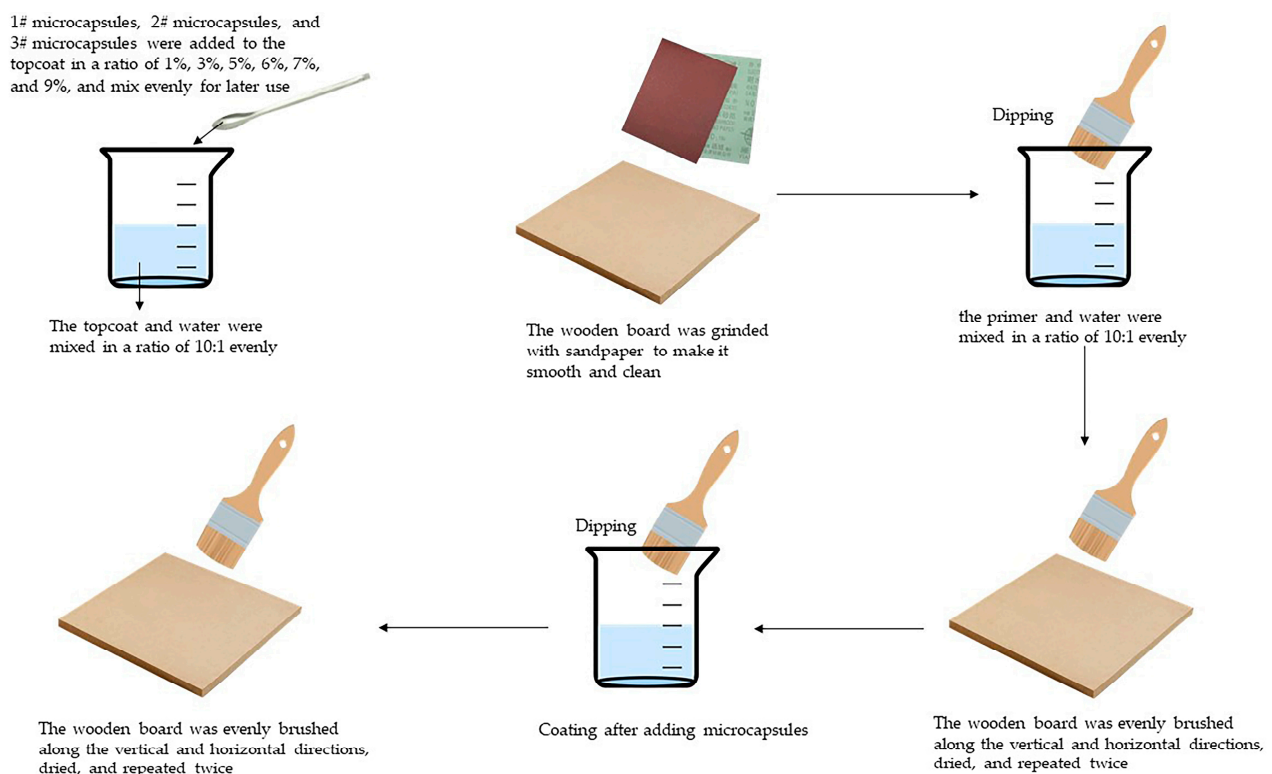


Figure 1. Coating process.

Table 5. List of materials for coatings.

Content of the Microcapsules (%)	Microcapsule Weight (g)	Coating Weight (g)
0	0	1.00
1.0	0.01	0.99
3.0	0.03	0.97
5.0	0.05	0.95
6.0	0.06	0.94
7.0	0.07	0.93
9.0	0.09	0.91

The Dulux water-based primer and topcoat were mixed with clean water at a ratio of 10:1 (volume). Then, the aloe emodin microcapsules were added to the topcoat. The aloe emodin microcapsules, prepared by SDBS, OP-10 and TWEEN-80, were named 1# microcapsule, 2# microcapsule and 3# microcapsule. They were added to the water-based topcoat at different concentrations of 1.0%, 3.0%, 5.0%, 6.0%, 7.0% and 9.0%, respectively, while maintaining the total quality of the coating. The MDF was exposed to an environment with a room temperature of 26 °C and a relative humidity of 60.0% ± 5.0% for one week, so that the moisture content of the MDF was approximately 12.0%. First, a substrate treatment was performed. A 320-grade sandpaper was used to polish the MDF, which can remove the burrs on the surface, and the debris was removed with a brush. Then, the dust was wiped with a rag so as to achieve the purpose of leveling and cleaning. Afterward, the primer was applied evenly along the vertical and horizontal directions on the MDF. After the first coating, the coated MDF was dried at room temperature for 20 min, then dried in a 40 °C oven for 20 min. After removal from the oven, the coated MDF was dried at room temperature for 4 h. The MDF was lightly polished with 600-grit sandpaper, and then the second coat of primer was applied, dried, and polished. The aloe emodin microcapsules and water-based topcoat were mixed evenly before applying the first coat of topcoat. The drying process was the same as above. After drying, the surface of the coating was lightly polished with 800-grit sandpaper, and then the second coat of topcoat was applied. In addition, the water-based coating without microcapsules was coated on the MDF as a blank control group.

2.4. Testing and Characterization

2.4.1. Microscopic Morphology and Chemical Composition

A scanning electron microscope (SEM, Thermo Fisher Scientific, Waltham, MA, USA) was used to observe the coatings. When using the SEM, the prepared coating was pasted on the sample table. Then, the sample was put into a specific position after gold spraying. In order to observe and record the micro-morphology of the sample, the observation multiple and focal length were adjusted to make the image reach the best state.

The chemical composition of the coating was analyzed using a Vertex 80V Fourier transform infrared spectroscopy (FTIR) instrument produced by Shanghai Smio Analytical Instrument Co., Ltd., Shanghai, China. For the infrared test, the infrared spectrum of the coating was measured using an attenuated total reflection device. The samples were placed on a test bench and fixed on the top surface of the pure diamond crystal using a pressure bar.

2.4.2. Antibacterial Properties

The antibacterial tests were carried out with *Escherichia coli* and *Staphylococcus aureus* as the experimental objects. According to GB/T 21866-2008 [35], the antibacterial tests were conducted on the coating. First, the nutrient agar medium powder was heated and dissolved in the purified water. Then, the mixture was divided into petri dishes to prepare planar nutrient agar media. The nutritional broth powder was added to the purified water and dissolved by heating to make a broth culture solution. An eluate with a concentration of 0.85% was obtained by dissolving NaCl in the purified water. The planar nutrient

agar medium, broth culture solution and eluent were heated for 30 min at 121 °C. A polyethylene film was immersed in a 70.0% alcohol solution for 30 min, washed with the eluent, and dried.

The bacteria on the slanted culture medium were transferred to the planar nutrient agar mediums using the sterilized inoculation ring. The planar nutrient agar mediums were then incubated in a constant temperature and humidity chamber at 37 °C for 20 h. Then, the fresh bacteria scraped 1–2 rings with the inoculation rings were added to the broth culture solution and mixed well. According to GB/T 4789.2-2022 [36], dilutions with tenfold increases in concentration were prepared sequentially to produce a 1:1000 bacterial suspension. A total of 0.5 mL of the bacterial suspension was dropped onto the surface of the prepared coating. The polyethylene films were picked up with forceps and laid flat on the coating so that the bacterial suspension was uniformly in contact with the coating. The coatings were put in petri dishes. The coatings were cultivated for 24 h in a constant temperature chamber with a temperature of 37 °C and a humidity of 98.0%. Each sample was tested in parallel with 2 sets. After 24 h of incubation, the coatings and polyethylene films were repeatedly rinsed with 20 mL of eluent, respectively. After adequate stirring of the eluent, 0.5 mL of the eluent was inoculated onto a planar nutrient agar medium. The planar nutrient agar medium was then incubated at a constant temperature of 37 °C and a humidity of 98.0% for 48 h in a controlled environment.

The planar nutrient agar medium, which had been incubated for 48 h, was removed and placed in the colony counter. The colonies on the bottom of the medium were observed and counted with a pen. The displayed number represented the total number of colonies in the planar nutrient agar medium. Two parallel trials were conducted, and the average of these two counts was considered the number of colonies for the sample. The actual number of viable bacteria extracted from each sample after 48 h was obtained by multiplying the colony count result by 1000. The antibacterial rate of the coating was calculated using the following Formula (1). In the formula, R represents the antibacterial rate, B represents the actual number of viable bacteria of the coating without microcapsules recovered after 48 h, and C represents the actual number of viable bacteria of coating with microcapsules recovered after 48 h, in CFU/tablet.

$$R = \frac{B - C}{B} \times 100\%. \quad (1)$$

2.4.3. Optical Performance

(1) Chromatic aberration: According to GB/T 11186.3-1989 [37], the chromatic aberration of the coatings was measured using a hand-held colorimeter (Shanghai Hechen Energy Technology Co., Ltd., Shanghai, China). After calibrating the colorimeter, the coatings were tested and the L , a , b values were recorded. The L indicates brightness, and the higher the L value, the brighter the color. The a value indicates color from red to green: a positive value suggests a reddish color, and a negative value represents a greenish color. The b value indicates color from yellow to blue. A positive b value represents a yellowish color, and a negative b value represents a bluish color. The values of coating without a microcapsule are L_1 , a_1 , b_1 , and the values of coating with microcapsules are L_2 , a_2 , b_2 . Chromatic aberration ΔE was calculated according to Equation (2), where $\Delta L = L_2 - L_1$, $\Delta a = a_2 - a_1$, and $\Delta b = b_2 - b_1$.

$$\Delta E = \left[(\Delta L)^2 + (\Delta a)^2 + (\Delta b)^2 \right]^{\frac{1}{2}}. \quad (2)$$

(2) Gloss: According to GB/T 4893.6-2013 [38], the gloss of the coating was tested. The gloss meter was calibrated and the coating was tested at incidence angles of 20°, 60° and 85°, respectively. According to Equation (3), the light loss rate of the coating at an angle

of incidence of 60° was calculated. G_L is the light loss rate, G_0 is the gloss of the coating without microcapsules, and G_1 is the gloss of the coating with microcapsules.

$$G_L = (G_0 - G_1)/G_0 \times 100\%. \quad (3)$$

2.4.4. Mechanical Property

(1) Hardness: According to standard 20211076-T-606 [39], a man-carried coating hardness tester was utilized to test the hardness of the water-based coating. The pencil hardness that scratches the surface of the coating is the surface hardness of the coating.

(2) Adhesion: According to standard GB/T 4893.4-2013 [40], the adhesion of the water-based coatings was tested using a film scribe (Zhejiang Airuipu Instrument Co., Ltd., Quzhou, China). The coated MDF was placed on the operating table, a scoring knife was placed perpendicular to the surface of the coated MDF, and the surface of coated MDF was cut with a constant speed and force. The coated MDF was then rotated at 90° and the previous process was repeated on the cut to form a grid on the surface of the coated MDF. In this process, all cuts must run through the coated MDF. An adhesive tape was stuck onto the grid. After tearing off, the adhesion of the water-based coating was determined by measuring the peeling off of the coating.

(3) Impact resistance: According to standard GB/T 4893.9-2013 [41], the impact resistance of the water-based coating was tested using a coating impact tester. The maximum height of the impact block where the coating does not crack, that is, the impact strength, was measured.

2.4.5. Roughness

The surface roughness of the water-based coating can be obtained by placing a glass plate coated with the coating on a test bench and regulating the position of the contact pin until it contacts the coating.

3. Results and Discussion

3.1. MDF Surface Topography Analysis

Figures 2–4 show the macroscopic morphology of MDF with different levels of 1# microcapsule, 2# microcapsule and 3# microcapsule, respectively. As the MDF itself was dark and brownish-yellow in color and the water-based coating was transparent, the three groups of MDF had similar surface colors from a macroscopic point of view. The addition of the three yellowish microcapsules had little influence on the surface color of the MDF. As can be seen from Figure 4F, when the microcapsule content was 9.0%, the surface of the water-based coating showed obvious particles and uneven coating. This is because at higher levels of microcapsules, the topcoat becomes denser and less fluid, resulting in uneven distribution of microcapsules and partial accumulation.

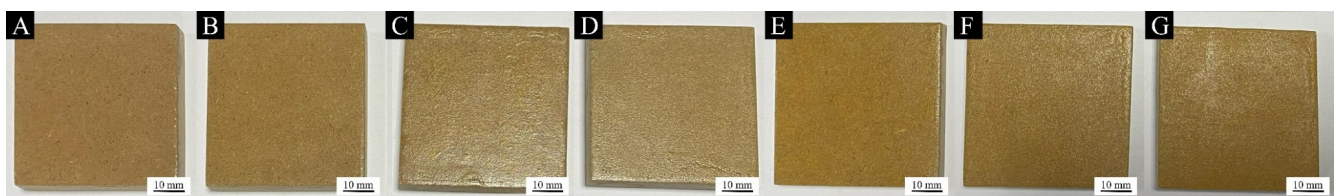


Figure 2. Macro-morphology of coating with different contents of 1# microcapsule on the MDF: (A) 0%, (B) 1.0%, (C) 3.0%, (D) 5.0%, (E) 6.0%, (F) 7.0%, (G) 9.0%.

Figure 5 shows the surface microstructure of a water-based coating without microcapsules and a water-based coating with 7.0% of 1# microcapsule, 2# microcapsule and 3# microcapsule, respectively. The surface of the water-based coating without the addition of microcapsules was relatively smooth. But the surface of the water-based coating with the addition of microcapsules had aggregated particulate matter, which was caused by the

presence of microcapsules in the coating in a particulate state. The water-based coating with 7.0% of the 3# microcapsule had the most particulate matter. This is because the 3# microcapsule was more heavily aggregated and adhered. Therefore, the aloe emodin microcapsules were unable to disperse evenly after being added to the water-based coating and applied to the surface of the MDF.

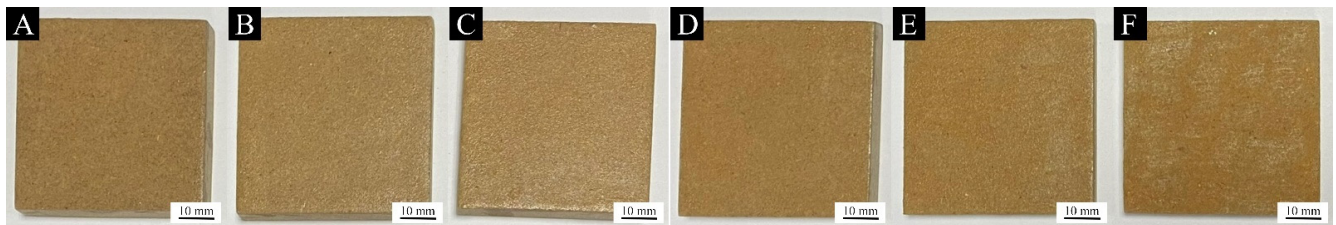


Figure 3. Macro-morphology of coating with different contents of 2# microcapsule on the MDF: (A) 1.0%, (B) 3.0%, (C) 5.0%, (D) 6.0%, (E) 7.0%, (F) 9.0%.

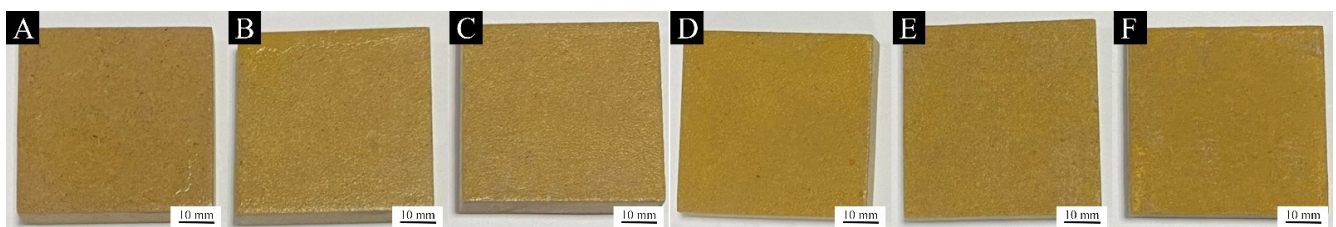


Figure 4. Macro-morphology of coating with different contents of 3# microcapsule on the MDF: (A) 1.0%, (B) 3.0%, (C) 5.0%, (D) 6.0%, (E) 7.0%, (F) 9.0%.

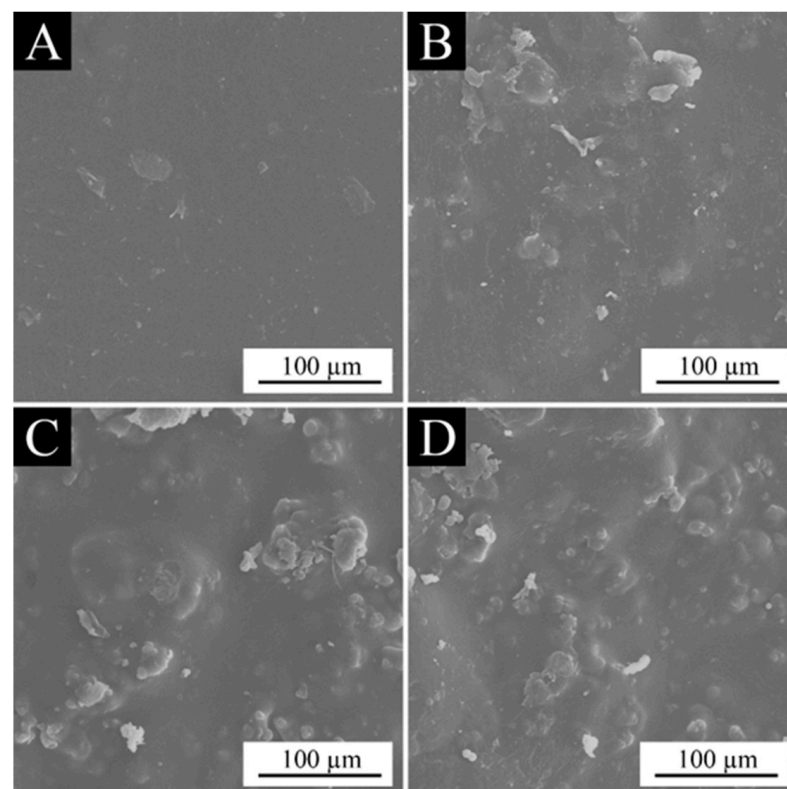


Figure 5. SEM images of coating on MDF: (A) without microcapsules, (B) with 7.0% of 1# microcapsule, (C) with 7.0% of 2# microcapsule, (D) with 7.0% of 3# microcapsule.

3.2. Chemical Composition Analysis of Coatings

The infrared spectra of a water-based coating of blank control group and the water-based coating with three kinds of the microcapsules are displayed in Figure 6. The chemical groups that cause various characteristic peaks are shown in Table 6. Infrared spectra analysis revealed the asymmetric C–H (sp^3) stretching vibration peak and the C–O vibration peak at 2950 and 1144 cm^{-1} , appearing in both water-based coatings and microcapsules, which indicates the presence of these functional groups in the coating. The characteristic peak at 1730 cm^{-1} belongs to the asymmetric stretching vibration peak of C=O in water-based coatings. These characteristic peaks were visible in all four curves, indicating that the chemical composition of the water-based coating was not destroyed after mixing with the microcapsules. The absorption peak at 1247 cm^{-1} was attributed to the contraction vibration of C–N and the deformation vibration of N–H in urea–formaldehyde resin. The peak at 1380 cm^{-1} comes from aryl–O–H bending vibration. The peak at 1450 cm^{-1} was the bending vibration of C–H with CH_2 or CH_3 groups. At 1560 cm^{-1} , the absorption peak corresponding to C=C in the microcapsule core of aloe emodin was visible in the coating curve, which confirmed that the chemical composition of the microcapsules remained intact in the water-based coating. To sum up, the presence of characteristic peaks in all curves indicated that no chemical reaction was caused by the addition of microcapsules in the water-based coating, ensuring the stability of the chemical composition of the coating. After being applied to the surface of MDF, the compositions of both water-based coating and microcapsules retained their original composition and effectiveness.

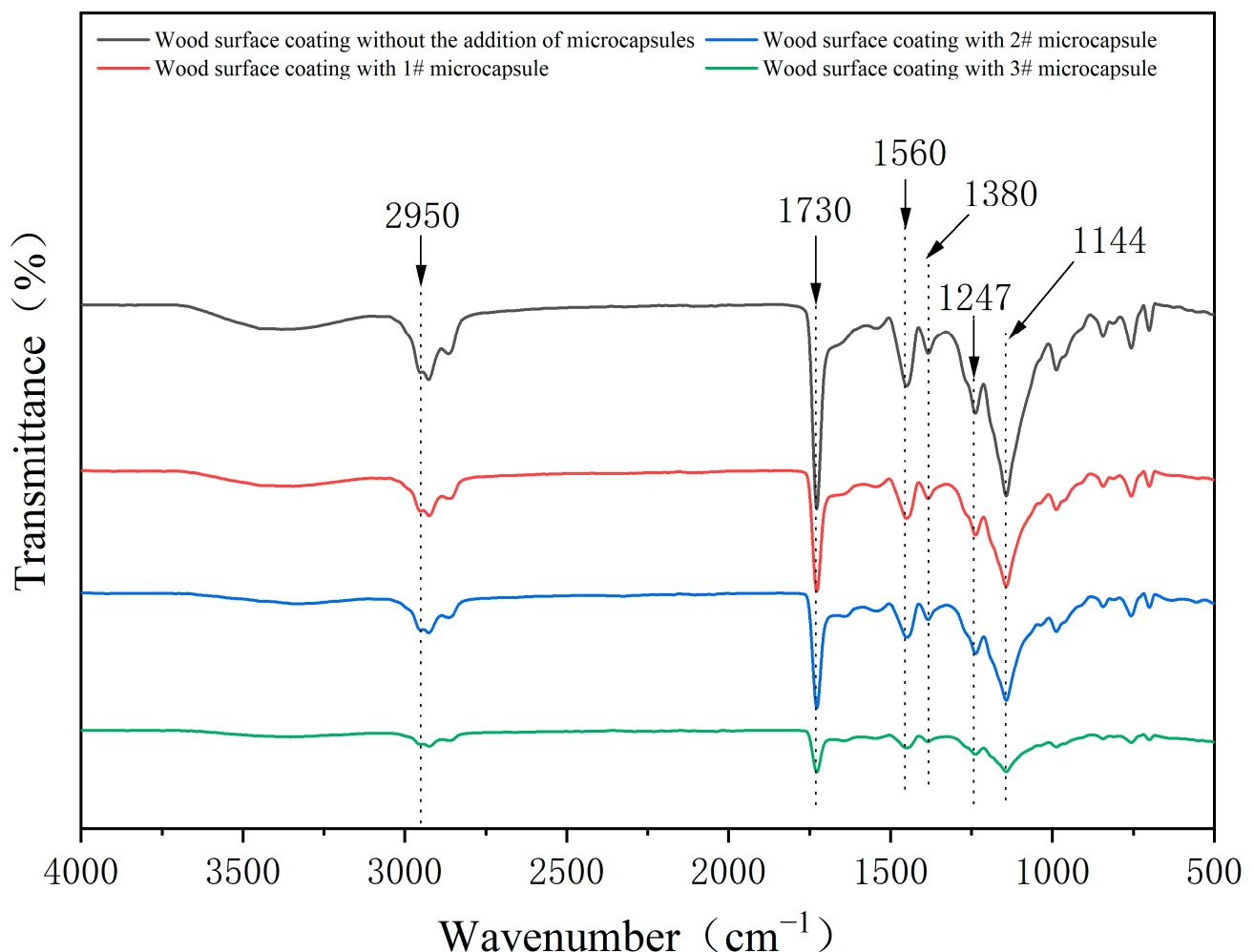


Figure 6. FTIR image of the coatings on the surface of MDF.

Table 6. Chemical groups leading to various characteristic peaks.

Characteristic Peak (cm ⁻¹)	Chemical Groups
2950	C-H (sp ³)
1144	C-O
1730	C=O
1247	C-N, or N-H
1380	aryl-O-H
1450	C-H with CH ₂ or CH ₃

3.3. Analysis of Antibacterial Performance of Coating on MDF

Theoretically, a higher content of aloe emodin microcapsules in the coating leads to a stronger antibacterial effect of aloe emodin. This hypothesis is confirmed by the results presented in Figure 7 and Table 7. The results revealed the number of recovered colonies and the antibacterial rate after the antibacterial tests of three different water-based coatings added with different microcapsules against *Escherichia coli* and *Staphylococcus aureus*. Interestingly, the antibacterial effect of the three groups of water-based coating against *Escherichia coli* was slightly better than that against *Staphylococcus aureus*. Compared to the coating of the blank control group, the antibacterial rates of the three groups of samples against both bacteria gradually increased, and the antibacterial effect also gradually enhanced. When the addition amount of microcapsules was between 3.0% and 6.0%, the antibacterial effect of microcapsules improved quickly. When the addition amount was between 6.0% and 9.0%, the improvement of antibacterial rate was relatively smooth. The water-based coating on the MDF added with the 1# microcapsule had the highest antibacterial rates of 74.1% and 66.0% against two bacteria, respectively. The water-based coating on the MDF added with the 2# microcapsule had the highest antibacterial rates of 70.0% and 62.8% against two bacteria, respectively, and the water-based coating on the MDF added with the 3# microcapsule had the highest antibacterial rates of 67.0% and 61.9% against two bacteria, respectively. The antibacterial effect of water-based coating on the MDF with the 1# microcapsule for *Escherichia coli* and *Staphylococcus aureus* was slightly stronger than that of the other two groups. These results indicated that the addition of microcapsules enhanced the antibacterial rate of water-based coatings on the surface of the MDF, and antibacterial effects were successfully exerted on the surface of the MDF. Because the surface of the MDF is not smooth and flat, with some small bumps and bumps, this not only provides a certain space for bacterial growth, but also makes the distribution of microcapsules less uniform. Therefore, the antibacterial effect of microcapsules in water-based coating on MDF is not particularly high.

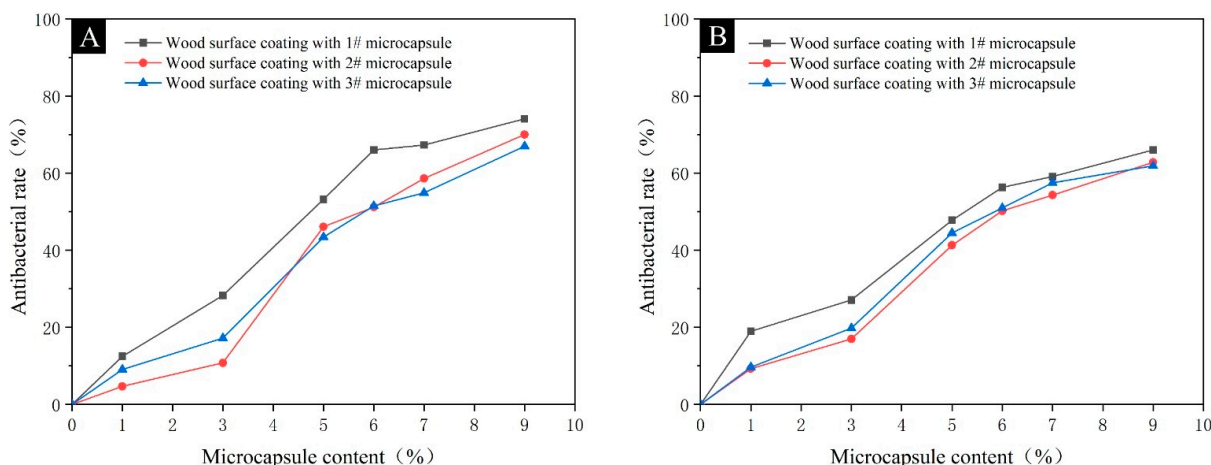
**Figure 7.** Antibacterial rate of MDF surface: (A) against *Escherichia coli*, (B) against *Staphylococcus aureus*.

Table 7. Actual number of viable bacteria and antibacterial rate of water-based coating.

Coating	Microcapsule Content (%)	Actual Number of Viable Bacteria for <i>Escherichia coli</i> (CFU·Tablets ⁻¹)	Antibacterial Rate for <i>Escherichia coli</i> (%)	Actual Number of Viable Bacteria for <i>Staphylococcus aureus</i> (CFU·Tablets ⁻¹)	Antibacterial Rate for <i>Staphylococcus aureus</i> (%)
1# Microcapsule Added to the Coating	0	297	-	247	-
	1.0	260	12.5	200	19.0
	3.0	213	28.3	180	27.1
	5.0	139	53.2	129	47.8
	6.0	101	66.0	108	56.3
	7.0	97	67.3	101	59.1
	9.0	77	74.1	84	66.0
2# Microcapsule Added to the Coating	0	297	-	247	-
	1.0	283	4.7	224	9.3
	3.0	265	10.8	205	17.0
	5.0	160	46.1	145	41.3
	6.0	145	51.2	123	50.2
	7.0	123	58.6	113	54.3
	9.0	89	70.0	92	62.8
3# Microcapsules Added to the Coating	0	297	-	247	-
	1.0	270	9.1	223	9.7
	3.0	246	17.2	198	19.8
	5.0	168	43.4	137	44.5
	6.0	144	51.5	121	51.0
	7.0	134	54.9	105	57.5
	9.0	98	67.0	94	61.9

3.4. Analysis of Optical Properties of Coating on MDF

The influences of different microcapsules with different contents on the chromatic aberration of water-based coating on MDF is shown in Table 8 and Figure 8. When the microcapsule content was between 1.0% and 6.0%, the chromatic aberration on the MDF gradually increased. However, when the content of aloe emodin microcapsules surpassed 7.0%, the increase in chromatic aberration on the surface of the MDF was more significant. This indicated that when the microcapsule content was higher than 7.0%, the impact on the surface color of the MDF was greater. The *b* value, representing the degree of yellow color, tended to increase as the microcapsule content increased because the aloe emodin microcapsules were yellow. The *L* value, representing lightness and darkness, and the *a* value, representing red and green values, were less variable. The maximum chromatic aberration of water-based coating with the 1# microcapsule was 10.95, the maximum chromatic aberrations of water-based coating with the 2# microcapsule was 7.21, and the maximum chromatic aberration of water-based coating with the 3# microcapsule was 14.35. Among the three groups of water-based coatings, the chromatic aberration of the coated MDF with different contents of the 2# microcapsule was small because the color of the 2# microcapsules, which was prepared with the OP-10 emulsifier, is the lightest, thus exerting the least impact on the surface color of the water-based coating.

Table 9 and Figure 9 show the gloss and light loss rate of the water-based coating on the surface of MDF with different contents of microcapsules. The gloss of the water-based coating was negatively correlated with the amount of microcapsules. Figure 9 shows that when the content of microcapsules was between 0% and 5.0%, the gloss of the water-based coating decreased rapidly at a 60° incidence angle, and the light loss rate ascended rapidly. When the content of aloe emodin microcapsules was between 6.0% and 9.0%, the gloss approached 0 GU and the light loss rate approached 100.0%, indicating that the addition of microcapsules had a significant impact on the gloss of the water-based coating. Because the microcapsules are granular and the coating with the added microcapsules is considerably less flat after drying, the diffuse reflection phenomenon on the surface of the water-based coating is exacerbated, resulting in low gloss. From Table 9, the gloss of the water-based coating with the addition of the 1# microcapsule was slightly higher than that of the other two groups. However, the gloss values and variation trends of the water-based coating in

the three groups were very similar, which indicated that the change in emulsifier type had a relatively small impact on the gloss of the water-based coating.

Table 8. Chromatic aberration of water-based coating with different contents of microcapsules.

Coating	Microcapsule Content (%)	<i>L</i>	<i>a</i>	<i>b</i>	ΔE
MDF with 1# Microcapsule Added to the Coating	0	48.66 ± 0.70	13.29 ± 1.00	25.59 ± 0.70	-
	1.0	49.45 ± 0.90	10.76 ± 0.90	29.65 ± 0.80	4.85
	3.0	49.48 ± 0.70	12.55 ± 1.00	33.62 ± 1.50	8.11
	5.0	49.18 ± 0.90	14.33 ± 0.30	33.87 ± 1.10	8.37
	6.0	49.75 ± 1.50	13.94 ± 0.20	33.87 ± 0.50	8.38
	7.0	48.42 ± 1.20	15.18 ± 0.40	34.52 ± 1.50	9.14
	9.0	52.64 ± 1.70	11.67 ± 0.20	35.66 ± 0.60	10.95
MDF with 2# Microcapsule Added to the Coating	0	48.66 ± 0.70	13.29 ± 1.00	25.59 ± 0.70	-
	1.0	48.95 ± 0.60	12.86 ± 0.90	25.30 ± 0.60	0.59
	3.0	50.57 ± 0.90	11.12 ± 0.40	29.23 ± 1.10	4.65
	5.0	52.15 ± 1.60	11.99 ± 0.40	29.63 ± 0.60	5.50
	6.0	51.28 ± 1.80	11.45 ± 0.80	30.38 ± 0.70	5.77
	7.0	51.60 ± 1.80	12.62 ± 0.20	30.84 ± 1.20	6.06
	9.0	52.68 ± 1.10	12.12 ± 0.50	31.45 ± 0.90	7.21
MDF with 3# Microcapsule Added to the Coating	0	48.66 ± 0.70	13.29 ± 1.00	25.59 ± 0.70	-
	1.0	50.61 ± 0.20	9.33 ± 0.30	30.01 ± 0.20	6.25
	3.0	50.13 ± 0.40	9.55 ± 0.30	31.79 ± 1.30	7.39
	5.0	51.11 ± 1.90	9.53 ± 0.40	32.38 ± 1.40	8.14
	6.0	52.43 ± 1.80	10.04 ± 0.10	33.56 ± 1.80	9.40
	7.0	55.21 ± 1.60	9.49 ± 0.50	36.01 ± 0.70	12.89
	9.0	55.49 ± 1.30	9.82 ± 0.30	37.72 ± 0.60	14.35

Table 9. Gloss and light loss rate of coating with different contents of microcapsules.

Coating	Microcapsule Content (%)	Gloss at 20° (GU)	Gloss at 60° (GU)	Gloss at 85° (GU)	Light Loss Rate (%)
MDF with 1# Microcapsule Added to the Coating	0	50.15 ± 1.30	82.15 ± 2.10	95.6 ± 1.40	-
	1.0	20.65 ± 0.30	48.70 ± 1.60	53.3 ± 1.90	40.7
	3.0	5.80 ± 0.50	21.70 ± 0.40	19.1 ± 1.10	73.6
	5.0	2.58 ± 0.20	12.38 ± 0.70	7.2 ± 0.30	84.9
	6.0	1.70 ± 0.20	8.88 ± 0.50	4.7 ± 0.50	89.2
	7.0	1.10 ± 0.10	5.85 ± 0.10	2.0 ± 0.10	92.9
	9.0	0.85	3.60 ± 0.20	1.0 ± 0.10	95.6
MDF with 2# Microcapsule Added to the Coating	0	50.15 ± 2.50	82.15 ± 1.60	95.6 ± 1.40	-
	1.0	18.23 ± 2.10	42.93 ± 1.10	45.7 ± 2.10	47.7
	3.0	4.05 ± 0.20	15.98 ± 0.60	9.6 ± 0.60	80.6
	5.0	1.10 ± 0.10	6.00 ± 0.20	1.9 ± 0.10	92.7
	6.0	0.98 ± 0.10	4.03 ± 0.20	0.8 ± 0.10	95.1
	7.0	0.75 ± 0.20	3.23 ± 0.10	0.8 ± 0.10	96.1
	9.0	0.63 ± 0.10	2.08 ± 0.10	0.6 ± 0.10	97.5
MDF with 3# Microcapsule Added to the Coating	0	50.15 ± 3.70	82.15 ± 1.60	95.6 ± 1.40	-
	1.0	16.10 ± 1.50	40.40 ± 1.20	44.1 ± 1.50	50.8
	3.0	2.23 ± 0.20	11.18 ± 0.30	5.50 ± 0.30	86.4
	5.0	0.90 ± 0.10	4.38 ± 0.10	1.00 ± 0.10	94.7
	6.0	0.68 ± 0.10	2.98 ± 0.10	0.70 ± 0.10	96.4
	7.0	0.68 ± 0.10	2.23	0.50 ± 0.20	97.3
	9.0	0.50 ± 0.10	1.80 ± 0.10	0.40 ± 0.10	97.8

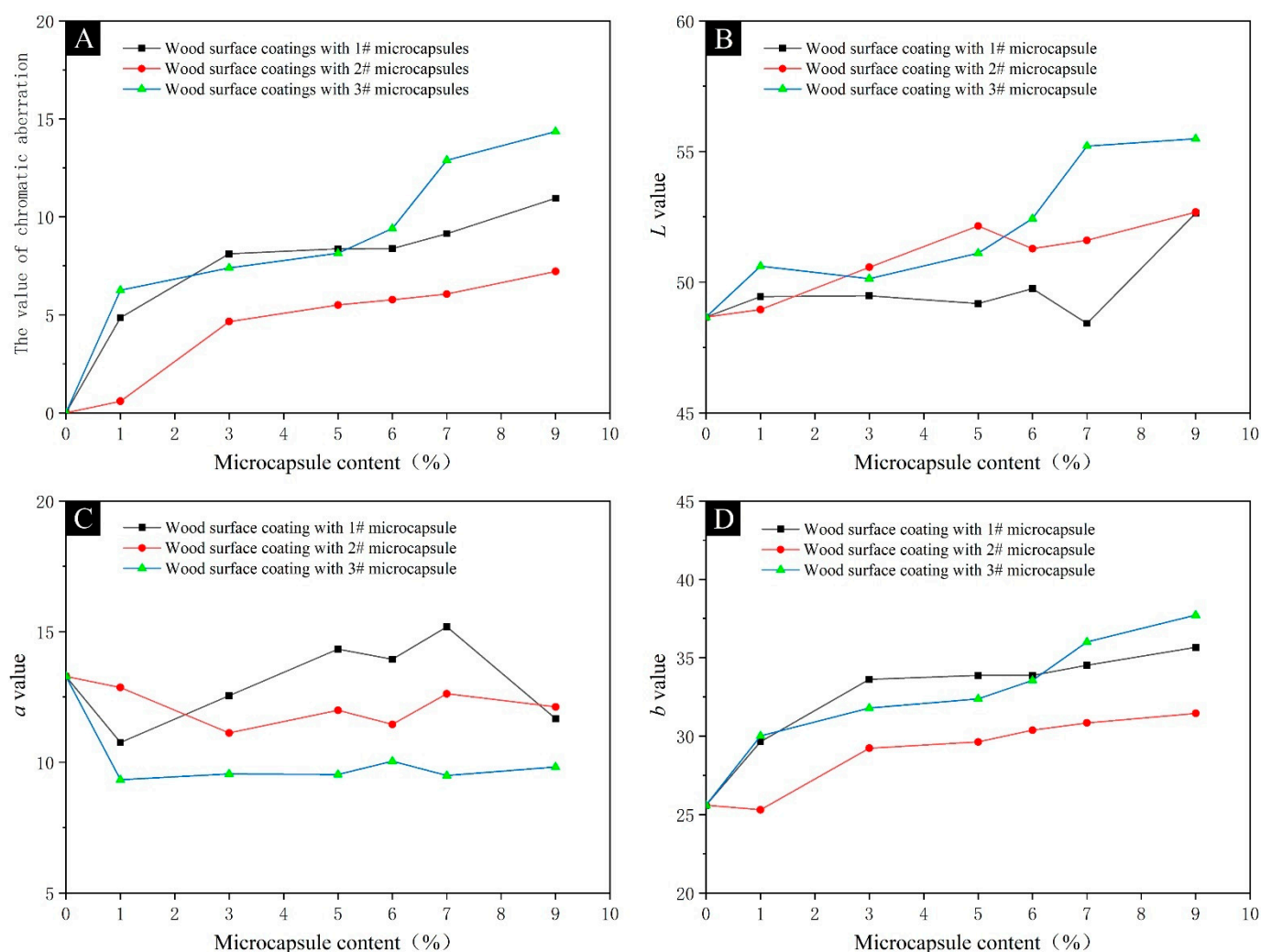


Figure 8. Effect of the microcapsule content on (A) chromatic aberration, (B) value of L , (C) value of a , and (D) value of b of the water-based coating.

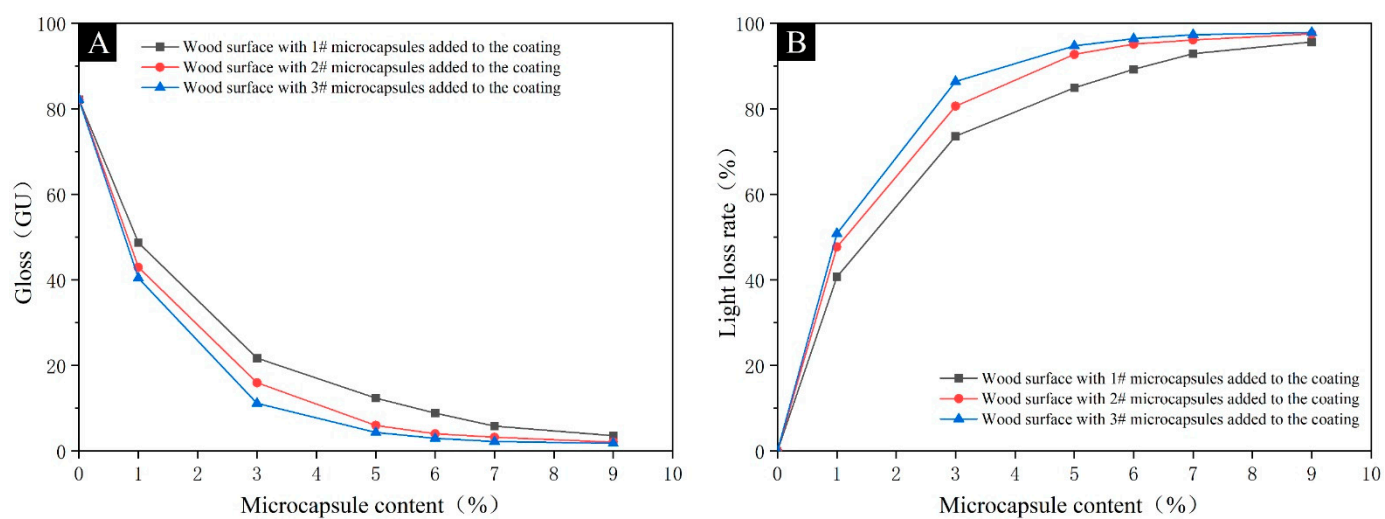


Figure 9. Effect of microcapsule content on gloss and light loss rate of water-based coating at 60°: (A) gloss, (B) light loss rate.

When exposed to light, a high-reflectivity coating reflected a certain range of light highly, preventing heat from accumulating on the surface of the water-based coating and increasing its temperature. This reduced the conduction of heat to the internal part of the MDF material, delaying the aging of the MDF after heating and extending the use life of the MDF and coating. Figure 10 reflects the reflectivity of water-based coating in the visible light band under different microcapsule kinds and different additive contents. As the content of microcapsules increased, the reflectivity of the three groups of water-based coating to visible light slightly increased. The trend of the three sets of reflectivity curves was basically consistent, and the spacing distance between the curves was very close. These indicated that the type of microcapsules prepared with different emulsifiers had a small impact on the visible light reflectance value of the water-based coating on MDF. The reflectivity curves of adding microcapsules in Figure 10A,C show a small absorption peak in the 450 nm wavelength range because the microcapsules themselves are yellow in color and absorbed a portion of the purple light at 450 nm, resulting in a slight decrease in reflectivity. Due to the light color of the 2# microcapsule, there is no obvious peak at 450 nm in Figure 10B.

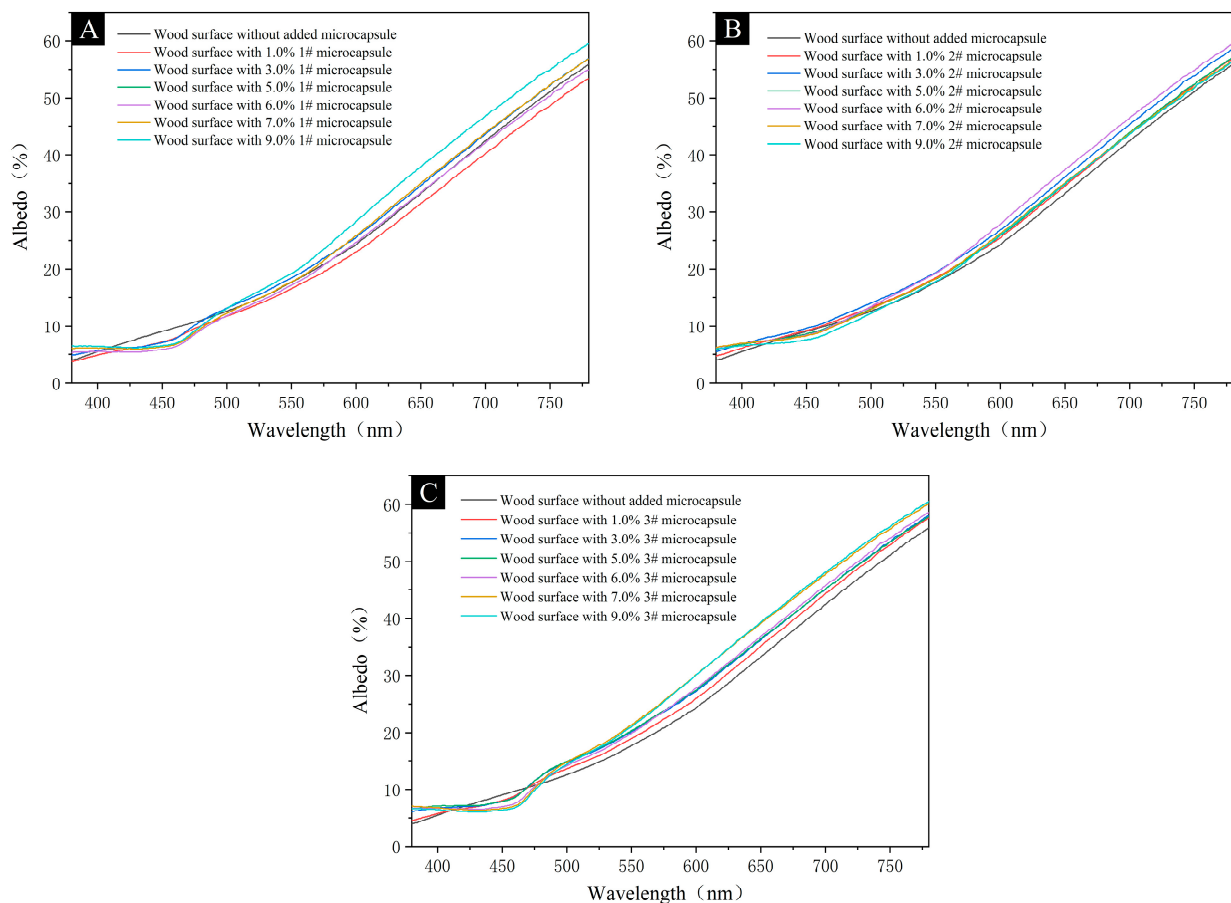


Figure 10. Reflectivity of visible light band of water-based coating: (A) with 1# microcapsule, (B) with 2# microcapsule, (C) with 3# microcapsule.

Based on the chromatic aberration, gloss, and visible light reflectivity of the three groups of water-based coatings, with the microcapsule content increasing, the chromatic aberration of the three groups of water-based coatings gradually increased. But the chromatic aberration values were all small. The gloss decreased significantly and the light loss rate increased. The reflectivity in the visible light band slightly increased. The changes in optical properties of the three groups of samples were similar, indicating that different types of emulsifiers had no significant effect on the optical properties of water-based

coating. Among them, MDF with the 2# microcapsule added to the coating has a smaller chromatic aberration, and the gloss of the water-based coating with the addition of the 1# microcapsule was the highest.

3.5. Analysis of Mechanical Properties of Coating on MDF

The effects of different microcapsules with different contents on the mechanical properties of water-based coating are shown in Table 10 and Figure 11. The coating was a water-based acrylic resin coating, which was self-crosslinking. Therefore, the water-based coating without microcapsules had a relatively soft hardness of HB because of the inherent characteristics of the water-based coating. With the rise of microcapsule content, the hardness of the water-based coating on the MDF improved from HB to 3H because the addition of microcapsules increases the density and hardness of the water-based coating. Among the three groups of samples, the coating with the 1# microcapsule exhibited the most significant increase in hardness. When the addition amount was 5.0%, the coating hardness already reached 3H. When the microcapsule content was between 0% and 7.0%, the adhesion level of the three coatings did not change, and all of them were level 1. This phenomenon indicated that when the microcapsule content was low, it did not influence the adhesion force of the water-based coating on the MDF. When the microcapsule content was 9.0%, the adhesion of all three coatings decreased slightly, and the adhesion level was 2. This is because when the coating contained more microcapsule particles, the content of adhesive in the coating dropped, leading to a decrease in adhesion of the water-based coating. Excessive microcapsules affected the adhesion of the coating to the substrate. When the microcapsule content was between 0% and 5.0%, the impact resistance of the three coatings did not change, and it was 4 kg·cm. When the content of aloe emodin microcapsules increased from 5.0% to 6.0%, the impact resistance of the three coatings ascended to 5 kg·cm. The aloe emodin microcapsules were particles with a “shell-core” structure, and the wall substance of the microcapsules played a shelter role on the key material. When the aloe emodin microcapsules were present in the coating, the numerous microcapsule particles generated a protective layer. The layer dispersed some of the force when subjected to impact, thereby improving the impact resistance of the coating.

Table 10. Mechanical properties of coatings on MDF with different contents of microcapsules.

Coating	Microcapsule Content (%)	Hardness	Adhesion (Grade)	Impact Resistance (kg·cm)	Roughness (μm)
MDF with 1# Microcapsule Added to the Coating	0	HB	1	4	0.37 ± 0.10
	1.0	H	1	4	0.79
	3.0	2H	1	4	2.10 ± 0.20
	5.0	3H	1	4	2.81 ± 0.30
	6.0	3H	1	5	2.92 ± 0.10
	7.0	3H	1	5	3.30 ± 0.10
	9.0	3H	2	5	3.90 ± 0.10
MDF with 2# Microcapsule Added to the Coating	0	HB	1	4	0.37 ± 0.10
	1.0	H	1	4	1.55
	3.0	2H	1	4	2.20 ± 0.10
	5.0	2H	1	4	3.32 ± 0.20
	6.0	2H	1	5	4.41 ± 0.10
	7.0	3H	1	5	4.84 ± 0.20
	9.0	3H	2	5	5.86 ± 0.10
MDF with 3# Microcapsule Added to the Coating	0	HB	1	4	0.37 ± 0.10
	1.0	2H	1	4	1.16 ± 0.20
	3.0	2H	1	4	3.09 ± 0.10
	5.0	2H	1	4	4.28 ± 0.10
	6.0	2H	1	5	4.75 ± 0.10
	7.0	3H	1	5	5.74
	9.0	3H	2	5	6.93 ± 0.10

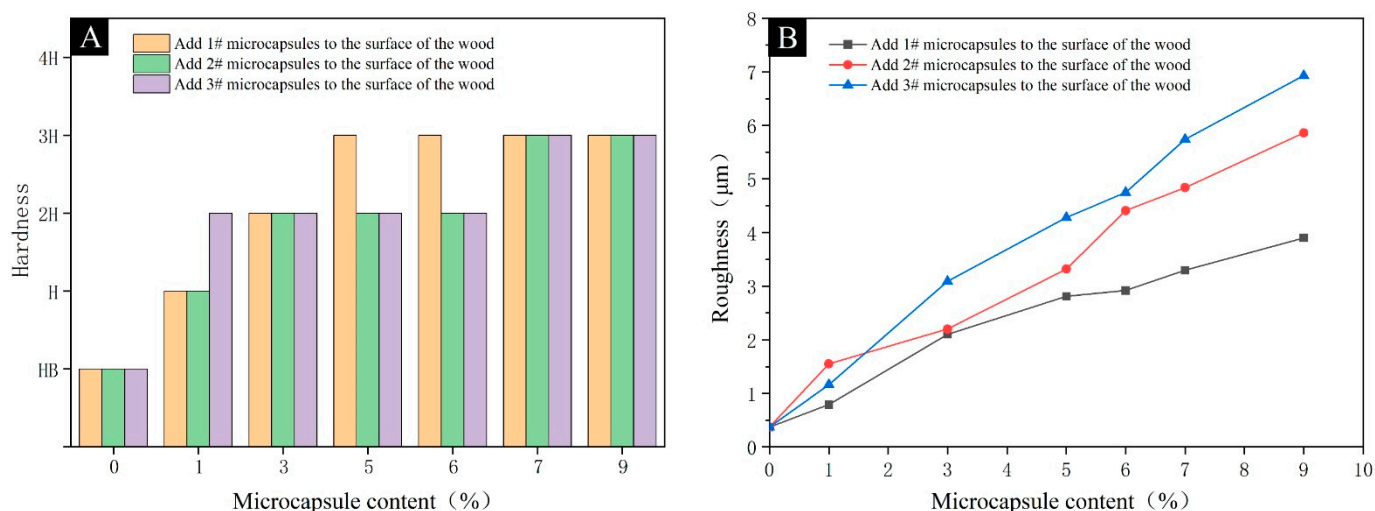


Figure 11. Effect of different contents on hardness and roughness of coating: (A) hardness, (B) roughness.

In Figure 11B, it can be seen that the surface roughness of the three groups of water-based coatings continuously increased with the rise of microcapsule content, resulting in rise in surface roughness. Microcapsules were spherical particles. After the coating was dried and cured, they still existed in the coating as granules and adhered to the surface of the MDF. Therefore, the flatness of the water-based coating reduced and became rough. Among them, the overall roughness of coating on the MDF with the addition of the 1# microcapsule was lower than that of the other two groups because the 1# microcapsule has the smallest average grain size among the three microcapsules and has the least agglomeration, which is dispersed more evenly in the coating. Therefore, the negative impact on the surface smoothness of the water-based coating was lower than that of the 2# microcapsule and the 3# microcapsule.

Based on the comprehensive analysis of the hardness, adhesion, impact resistance, and roughness of the water-based coatings on MDF, as the content of microcapsules increased, the hardness of the three coatings gradually rose. But the adhesion slightly decreased, the impact resistance slightly strengthened, and the roughness of the coatings increased. Among them, the coating on the MDF with the 1# microcapsule had a higher hardness and the lowest roughness, making it the group with the best mechanical properties among the three groups of water-based coatings with different microcapsules.

4. Conclusions

Three kinds of aloe emodin microcapsules prepared with SDBS, OP-10 and TWEEN-80 emulsifiers were incorporated into water-based coatings and applied to the surface of MDF. The morphology, chemical composition, antibacterial properties, optical properties, and mechanical properties of water-based coating were examined. The addition of microcapsules improved the antibacterial properties of the water-based coatings on the MDF surface. The antibacterial rates of the coatings increased with higher microcapsule content. The water-based coating with the 1# microcapsule prepared with SDBS had the highest antibacterial rates against other two types of OP-10 and TWEEN-80. When the microcapsule content was 9.0%, the antibacterial rates of water-based coating against *Escherichia coli* and *Staphylococcus aureus* reached 74.1% and 66.0%, respectively. The water-based coating with 2# microcapsules prepared with OP-10 had the antibacterial rates of 70.0% and 62.8% against *Escherichia coli* and *Staphylococcus aureus*, respectively. The water-based coating with 3# microcapsules prepared with TWEEN-80 had antibacterial rates of 67.0% and 61.9% against *Escherichia coli* and *Staphylococcus aureus*, respectively. As the microcapsule content increased, the surface chromatic aberration of the three groups of water-based coating slightly increased. The gloss level significantly decreased. The reflectivity of the visible band slightly increased. The adhesion of the water-based coating slightly decreased.

While the impact resistance increased, the hardness increased significantly. The roughness also increased. Based on the morphology, antibacterial properties, optical properties, and mechanical properties of the water-based coating on MDF surface, the water-based coating with the 1# microcapsule prepared with a 3.0% concentration of SDBS as an emulsifier had the best comprehensive performance. These findings provide technical references for utilizing aloe emodin microcapsules in antibacterial coatings on MDF surface, and its potential applications in coatings and other industries are expanded.

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